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Optimizing Water Quality Treatment Levels for Water Distribution Systems under Mixing Uncertainty at Junctions

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Abstract: A real-life water distribution system (WDS) contains uncertainty in numerous stages. This makes the optimal management and design of a WDS a complex problem. Water quality has also become a significant factor in the design and management of a WDS. Our objective was to incorporate water quality uncertainty in the WDS design problem. The mixing level was assumed to be uncertain and used to design the WDS such that the design was immune to the level of mixing. This method aimed to yield designs that satisfied the nodal concentration constraints irrespective of the mixing level in the junctions. Two optimization methodologies, robust optimization and info-gap decision theory combined with a cuckoo search optimization algorithm, were proposed to solve this problem. An illustrative example 4×4 grid network was used to understand nonuniform mixing and explain the design methodology using both methodologies. Then these methodologies were applied to solve a similar treatment plant problem on a modified Fossolo network. The results also exhibited a significant variation in cost between complete mixing and nonuniform mixing. The WDS designs obtained from both methods were evaluated through Monte Carlo simulations. **DOI: 10.1061/(ASCE)WR.1943-5452.0001544.** *© 2022 American Society of Civil Engineers.*

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Introduction

A water distribution system (WDS) is a network consisting of pipes, pumps, tanks, and valves that work in unison to provide sufficient quantities of high-quality water (clean, clear, and odorless) at proper pressures to downstream consumer taps. As populations grow and increasingly move toward urban centers, a more significant fraction of the world's population is becoming increasingly dependent on centralized WDSs for water supply. Hence, WDS design and operation must be managed appropriately. Since the Safe Drinking Water Act (SDWA) passage in 1990, researchers have been motivated to work on water quality and system hydraulic issues in WDS design and operation. Today, maintaining desired water quality (WQ) levels and pressure are equally important (Pasha and Lansey 2010). Other than a few quantities known to be precise, like the length of the pipes because its value remains constant over time (Gupta and Bhave 2007), most of the parameters involved in design of WDSs are subject to uncertainty. The uncertainty is caused either in parameter measurement, future estimation, or the assumptions made to simplify the model. The combined result of these uncertainties leads to erroneous estimates, resulting in reduced reliability of the design.

Uncertainty in Water Distribution Systems

Lansey et al. (1989) were the pioneers in designing WDSs under uncertain conditions. They worked with a probabilistic approach. They used chance-constraint formulation and solved using nonlinear programming techniques combined with generalized reduced gradient technique-II. They reported several designs and their corresponding reliability values for two-loop networks (TLNs). Kapelan et al. (2004) developed a method to obtain the least-cost designs for a desired level of reliability. The uncertain parameter was modeled as a probability density function, which was obtained using the Latin hypercube sampling (LHS) method. The least-cost problem was then solved using a genetic algorithm (GA) and a new robust GA, and the resulting designs were compared. Babayan et al. (2005) replaced Monte Carlo simulation (MCS) or any other sampling techniques by adding a margin of safety factor for pressure at nodes and assuming the fluctuations of heads at nodes are caused by uncertainty in demand. They then reformulated the stochastic formulation into a deterministic case and solved the design problem using GA. Babayan et al. (2007) further suggested two approaches to obtain the least-cost designs for a target system robustness. The first approach was an integration approach in which stochastic problem formulation was replaced with a deterministic one. Then, the optimization problem was solved with the standard GA. The second was the sampling approach (LHS), in which a GA solved the stochastic problem directly to evaluate each solution's fitness. Kang et al. (2009) studied alternative approximation techniques (first-order second moment and LHS), predicting the uncertain model with less computational effort.

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Multiobjective Optimization of WDS under Uncertainty

Tolson et al. (2004) developed a method combining GA and the first-order reliability method (FORM). FORM was used to compute the network capacity reliability of the design obtained in GA iterations; they considered both demand and roughness as uncertain and calculated reliability by estimating the probability it met the minimum pressure at nodes. Kapelan et al. (2005) proposed a multiobjective approach. The objectives were minimization of design cost and maximization of robustness (which is the probability to satisfy minimum pressures at all the nodes simultaneously). They considered future demands and roughness uncertain variables in their formulation. They used a popular nondominant sorting genetic algorithm-II multiobjective optimization algorithm. Unlike the previous methodologies considering only specific nodes, this method considered all nodes simultaneously. Giustolisi et al. (2009) proposed a two-stage optimization method to obtain the initial set of designs for a stochastic multiobjective optimization from the Pareto front of the deterministic multiobjective method. This method reduced the number of stochastic simulations, making it a computationally less expensive method. They used a multiobjective genetic algorithm to obtain Pareto fronts. They, too, considered both demands and roughness uncertain parameters. Jung et al. (2012) used multiobjective GA to solve the problem. The system's robustness was estimated as a disturbance index while considering the demand and roughness uncertain. They compared the results considering only critical nodes and considering all the nodes. They concluded that consideration of only critical nodes might lead to less robust designs. Basupi and Kapelan (2015) developed a methodology to obtain adaptable design solutions in the future for an existing WDS. They devised a multiobjective optimization method that simultaneously minimized total intervention cost and maximized future resilience. They considered demand uncertain for this problem. They studied the New York tunnel expansion problem. Marquez Calvo et al. (2019) obtained the optimal valve management considering uncertainty in demands to improve water quality using robust optimization. They did a multiobjective study minimizing the water age and number of valve closures.

Most of the research in uncertainty in WDSs has focused on the design of WDSs. Probabilistic methods were used to handle the uncertainty in the parameters (like demand and roughness). Higher computational time and assumption of the probability density function are the major drawbacks of probabilistic methods. Nonprobabilistic uncertainty handling techniques were developed to overcome these drawbacks.

Robust optimization (RO) and info-gap decision theory (IGDT) are examples of nonprobabilistic uncertainty handling techniques. Chung et al. (2009) implemented a RO approach to model uncertain water demand in municipal water distribution systems with hypothetical conditions. Perelman et al. (2013a, b) used a robust counterpart (RC) approach to solve water distribution system design with uncertain designs. Housh (2017) used a similar RC approach for a flood control design problem, where the author proposed an iterative linearization method to formulate the RC in every iteration. Ghelichi et al. (2018) and Naderi and Pishvaee (2017) also used RO formulations to handle uncertainties in the water resources domain and stated their advantages over the probabilistic approach. Hipel and Ben-Haim (1999) investigated the application of IGDT for sustainable water resource management with uncertain supply. Roach et al. (2015) compared RO and IGDT optimization methods in water resource management under severe uncertainty. Their study aimed at obtaining optimal intervention options to minimize the installation and operation cost while handling the uncertain demand and supply constraints. Both these methods have proven useful and simple to handle uncertainty in the water resources domain.

Most of the previous works in WDS design under uncertainty proposed designs that can handle uncertainty in demands, pipe roughness, head, Hazen–Williams coefficient, or a combination of them (i.e., almost all hydraulic parameters). However, there are not many studies that have addressed uncertainty in water quality parameters. Pasha and Lansey (2005, 2010) examined the effect of a few uncertain sources like reaction coefficient and pipe diameter in the system's water quality analysis. They concluded that the uncertain quantification function depends on location and relative distance from the contaminant source. They also stated that the bulk reaction coefficient has the most influence among all the water quality parameters studied. Hart et al. (2019) considered the effect of hydraulic and water quality uncertainty to model contamination transport throughout the system and obtain the locations for drinking water sampling. They studied uncertainty in demand, bulk reaction rate coefficient, contaminant injection location, and many more input parameters as uncertain in their study for two exemplar water distribution networks.

In addition to these water quality parameters, the uncertainty in mixing at cross-junctions can also affect the consumer nodes' water quality requirements. Ho et al. (2008) showed experimentally that the assumption of complete mixing at cross-junctions could result in erroneous concentration calculations at different consumer nodes. Song et al. (2009) also stated that the assumption of instantaneous complete mixing at the cross-junctions could result in erroneous results. Water quality modeling, assuming nonuniform mixing at junctions, complicates the problem. Many experimental studies have been conducted to develop approximate models that can model the uncertainty of mixing. Ho and Khalsa (2008) developed a bulk advective mixing (BAM) model for addressing the nonuniform mixing behavior at cross-junctions. They incorporated a mixing parameter $s \in [0, 1]$ that governs the extent of mixing, with incomplete mixing the lower bound and complete mixing the upper bound. Song et al. (2009) also developed a model named AZRED for modeling incomplete mixing at specific junctions. Webb and Van Bloemen Waanders (2007) modeled the mixing phenomenon using a computational fluid dynamics (CFD) model and showed its efficiency with experiments. There have been many studies in accurately modeling nonuniform mixing with CFD-based models. Yu et al. (2015) developed a nonuniform mixing model at junctions with different pipe diameters but the same diameters in opposite pipes. This model used the Reynolds number and other flow characteristics, making it complicated. There has not been any simple model to predict the mixing phenomenon at the junctions accurately. Paez et al. (2017) compared the water quality at consumer nodes using the complete mixing and BAM model for WDSs of two cities and two different cases of a grid network. They concluded that the mixing uncertainty is more significant in a grid-type network containing many cross-junctions than the city WDSs in the study.

The current design of WDSs does not consider various uncertainties in water quality affecting parameters. This assumption may lead to unsatisfactory designs, which may fail to satisfy the water quality constraints. The mixing at cross-junctions is assumed to be instantaneous and complete, which was proven wrong by Ho and Khalsa (2008) and Song et al. (2009). The exact mixing phenomenon is complex, and complex modeling software involving CFD simulation is required (Webb and Van Bloemen Waanders 2007). Because WDSs have multiple cross-junctions and using such complex solvers would take a lot of time to overcome this problem, it is necessary to obtain designs immune from any level of mixing.

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The current study aims to obtain robust treatment levels that are immune to any level of mixing within the desired uncertainty set.

This work is expected to contribute to two vaguely explored areas in the design of WDS. One area is to identify and quantify the significance of nonuniform mixing uncertainty in the design of WDSs. The other area is converting the complex design problem into a simplified convex optimization problem using RO and IGDT and then obtain the optimal WDS design for the aforementioned problem.

Methodology

It is difficult to quantify the exact level of mixing for given flow values arriving at a cross-junction. This problem only increases n-fold with the increase in cross-junctions in the network. Incorporating nonuniform mixing is completely new, and no studies have been done to estimate the probability distribution function of nonuniform mixings. RO and IGDT techniques can produce efficient solutions without the need for probabilistic distributions of uncertain variables. The current study used these two methods to obtain designs that ensure immunity against the unknown mixing parameters within an uncertainty set.

The BAM model developed by Ho et al. (2008) was used in the current study. Ho et al. (2008) showed that when two fluid streams with different flow values and concentrations enter a cross-junction, it will cause a bulk transfer of fluid and solutes between incoming and outgoing streams. Fig. 1 depicts the bulk mixing (incomplete mixing) at a cross-junction, where Q represents flow values and C represents concentration.

The bulk mixing equations for the case are shown in (Fig. 1) where $Q_1 > Q_2$ can be written easily with mass balance

$$C_4 = C_1 \tag{1}$$

$$C_3 = \frac{Q_2 C_2 + (Q_1 - Q_4) C_1}{Q_3} \tag{2}$$

whereas complete mixing equations are

$$C_3 = C_4 = \frac{Q_1 C_1 + Q_2 C_2}{Q_3 + Q_4} = \frac{Q_1 C_1 + Q_2 C_2}{Q_1 + Q_2} \tag{3}$$

Ho et al. (2008) stated that the real-life scenario is somewhere between bulk and complete mixing, with the final concentration as given in Eq. (4)

$$C_{\text{scaled}} = C_{\text{bulk}} + s(C_{\text{complete}} - C_{\text{bulk}}) \tag{4}$$

where s = mixing parameter. The mixing parameter s is assumed to be an uncertain variable in this work.

The explicit equations for concentration at various demand nodes considering s an uncertain variable based on these BAM equations get complicated with an increase in the number of cross-junctions. In order to simplify these explicit equations, a data-driven linear surrogate model was developed for a specific set of flows in the network under steady-state assumptions for each demand node. The equation obtained through linear regression for a demand node j is of the form shown in Eq. (5)

$$C^j_{\text{out}}(\text{CR}, Q, \mathbf{s}(\xi))_{\text{BAM}} = \mathbf{a}^{\mathbf{T}}_{\mathbf{j}}\text{CR} + \text{CR}^T \mathbf{E}_j \mathbf{s} \tag{5}$$

where C^j_{out} = outlet concentration; $\mathbf{a_j}$ = vector; $\mathbf{E_j}$ = matrix, all obtained from regression modeling; CR = reservoir concentration after treatment; and $\mathbf{s}$ = uncertain mixing parameter vector.

We used a linear regression model so that the RO and IGDT reformulations were easier to exhibit. As far as we know, any convex approximation model should work with the RO and IGDT.

Treatment Cost Optimization

A treatment plant is used to treat the water from the reservoir before entering the WDS. The uncertainty in mixing affects the treatment cost. The objective of this problem is to minimize the treatment level cost. Treatment plants are designed for existing water distribution networks. This simplification makes it a simpler problem that does not deal with hydraulics. Because the hydraulic design is already done and the flow characteristics are assumed to be constant under steady-state conditions, the objective function is only

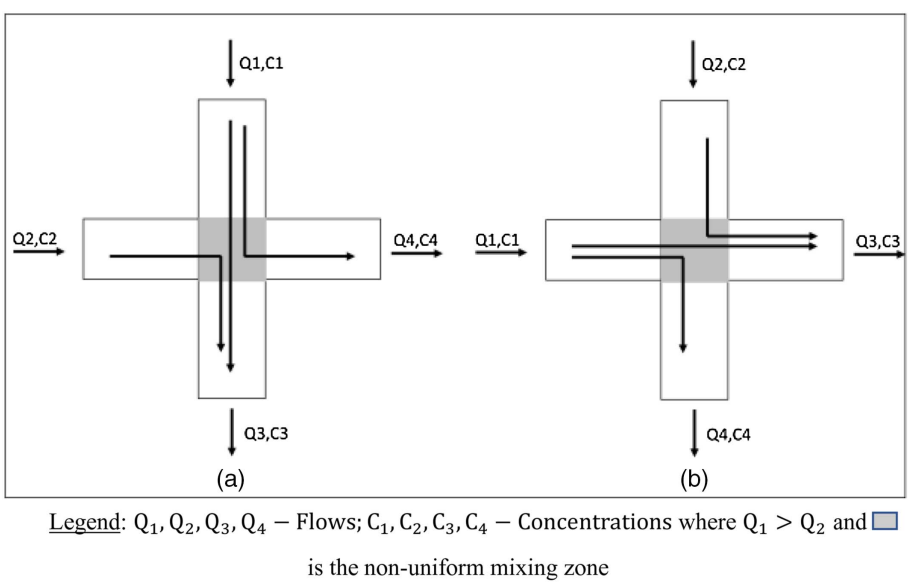

Fig. 1. Pictorial representation of incomplete mixing cases: (a) highest momentum is in the vertical direction; and (b) highest momentum is in the horizontal direction. (Adapted from Ho et al. 2008.)

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affected by nonuniform mixing at junctions. The problem structure can be written as follows:

$$\min \mathrm{TC}(\mathrm{ICR}, \mathrm{CR}) \tag{6}$$

$$\text{such that } C_{\mathrm{LL}}^{j} \le C_{\mathrm{out}}^{j}(\mathrm{CR}, Q, \mathbf{s}(\xi))_{\mathrm{BAM}} \le C_{\mathrm{UL}}^{j}, \quad \forall\, j \in J,\ \xi \in U \tag{7}$$

where TC = treatment cost; ICR = initial reservoir concentration before treatment; CR = concentration after the treatment; C_{LL}^{j}, C_{UL}^{j} = lower and upper limits of desired outlet concentration at junction j; and C_{out}^{j} = outlet concentration at node j.

Optimization under Uncertainty

Consider the following optimization problem denoted $\mathcal{P}$

$$\min_x f(x, \xi) \tag{8}$$

$$\text{such that } g(x, \xi) \le 0 \tag{9}$$

where x = decision variable; and ξ = parameter of the problem. If ξ is uncertain, then the exact value of ξ is unknown, making it difficult to optimize such a problem.

If ξ is known to be within a particular uncertainty set, the RO technique that can reformulate the uncertain problem to a tractable version called the RC (Ben-Tal and Nemirovski 1998). The RC can be solved to obtain a robust solution.

Another approach is the IGDT. It is useful in dealing with decision-making under severe uncertainty (Ben-Haim 2006). The result of the IGDT is an uncertainty set that the decision-maker is willing to tolerate. The higher the budget the decision-maker is willing to spend, the higher the uncertainty the system can tolerate.

In RO, the uncertainty set should be known to start the design problem, and we get the optimal robust design as the result. In IGDT, the designer should input a budget that he is willing to expend to get the uncertainty set (U) as well as a design that is immune to the uncertain variables in U as a result of the problem. Both these approaches are ways of dealing with different kinds of uncertainty.

Problem Formulation

The formulation [Eqs. (10)–(12)] is called the nominal problem denoted by $\mathcal{NP}$ and considers s a certain parameter and J the set of all nodes where demand is supplied

$$\min_{\mathrm{CR}} \mathrm{co}^T(\mathrm{ICR} - \mathrm{CR}) \tag{10}$$

$$\text{such that } C_{\min}^{j} \le a_j^T \mathrm{CR} + \mathrm{CR}^T E_j s \le C_{\max}^{j}, \quad \forall\, j \in J \tag{11}$$

$$0 \le \mathrm{CR} \le \mathrm{ICR} \tag{12}$$

where ICR = initial reservoir concentration before treatment; CR = reservoir concentration after treatment; and co = cost of unit treatment of water per unit volume.

Suppose s is considered an uncertain parameter. In that case, the constraint in Eq. (11) becomes Eq. (13), which should satisfy every realization of the parameter $s \in U$, which leads to an infinite number of constraints. The problem then becomes a semiinfinite optimization problem that is intractable

$$C_{\min}^{j} \le a_j^T \mathrm{CR} + \mathrm{CR}^T E_j s \le C_{\max}^{j}, \quad \forall\, j \in J,\ \forall\, s \in U \tag{13}$$

Robust Optimization

The RO approach for dealing with uncertainty in optimization is a worst-case–scenario approach (Housh et al. 2011). Consider the problem $\mathcal{P}$. Assume that some information about ξ is known. For example, assume that ξ lies in the polyhedral set U, where $U = \{\xi \in \mathbb{R}^n : \mathbf{A}\xi \le \mathbf{b}\}$ where $\mathbf{A}$ is a matrix and $\mathbf{b}$ is a vector. Then the problem now becomes

$$\min_x f(x, \xi) \tag{14}$$

$$\text{such that } g(x, \xi) \le 0, \quad \forall\, \xi \in U \tag{15}$$

The constraint should be satisfied for all possible realizations of ξ. This uncertain version of the problem is not tractable because there are infinite constraints [Eq. (15)]. RO uses the worst-case–scenario approach and dualization to convert this uncertain problem to a tractable robust counterpart (Ben-Tal and Nemirovski 1998).

RC Formulation

For this study, the uncertainty set in which s lies is assumed to be a polyhedral set; that is, $U = \{s \in \mathbb{R}^I : 0 \le s \le 1\}$, which can be rewritten as, $U = \{s \in \mathbb{R}^I : \mathbf{A}s \le \mathbf{b}\}$, where $\mathbf{A} = \begin{bmatrix} I^{I\times I} \\ -I^{I\times I} \end{bmatrix}$; $\mathbf{b} = \begin{pmatrix} e \\ 0 \end{pmatrix}$; $\mathbf{I}$ is the identity matrix; $\mathbf{e}$ is the vector of all 1; and 0 is the vector of all zeroes. The uncertain version of our problem can be reformulated using the following worst-case–scenario approaches:

$$\min_{\mathrm{CR}} \mathrm{co}^T(\mathrm{ICR} - \mathrm{CR}) \tag{16}$$

$$\text{such that } C_{\min}^{j} \le \min_{s\in U} a_j^T \mathrm{CR} + \mathrm{CR}^T E_j s, \quad \forall\, j \in J \tag{17}$$

$$\max_{s\in U} a_j^T \mathrm{CR} + \mathrm{CR}^T E_j s \le C_{\max}^{j}, \quad \forall\, j \in J \tag{18}$$

$$0 \le \mathrm{CR} \le \mathrm{ICR} \tag{19}$$

Now, the problem can be reformulated. The reformulated problem will satisfy the constraints for all possible realizations of $s \in U$ and is performed using dualization. Consider the minimization in the first constraint for some $j \in J$, which is equivalent to

$$\min_{s\in\mathbb{R}^I} a_j^T \mathrm{CR} + \mathrm{CR}^T E_j s \tag{20}$$

$$\text{such that } As \le b \tag{21}$$

Writing the Lagrangian and minimizing, the dual objective is

$$\min_{s\in\mathbb{R}^I} a_j^T \mathrm{CR} + \mathrm{CR}^T E_j s + \mu_j^T (As - b) \tag{22}$$

$$= \begin{cases} a_j^T \mathrm{CR} - \mu_j^T b, & \text{if } \mathrm{CR}^T E_j + \mu_j^T A = 0 \\ -\infty, & \text{else} \end{cases} \tag{23}$$

Using strong duality

$$\begin{aligned} C_{\min}^{j} &\le \min_{s\in U} a_j^T \mathrm{CR} + \mathrm{CR}^T E_j s \\ &= \max_{\mu\in\mathbb{R}_+^{2I}} a_j^T \mathrm{CR} - \mu_j^T b : \mathrm{CR}^T E_j + \mu_j^T A = 0 \end{aligned} \tag{24}$$

Then, by asserting that because the maximum of the argument is at least $C_{\min}^{j}$, there should exist at least one value of the dual

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variable (μ_j) that makes the argument at least $C^j_{\min}$. We drop the maximization. Thus, the infinite number of constraints can be replaced with constraints [Eqs. (25)–(27)] for each $j \in J$

$$C^j_{\min} \le a_j^T \mathrm{CR} - \mu_j^T b \tag{25}$$

$$\mathrm{CR}^T E_j + \mu_j^T A = 0 \tag{26}$$

$$\mu_j \ge 0 \tag{27}$$

Similarly, the upper bound constraint (for some $j \in J$) can be reformulated to

$$a_j^T \mathrm{CR} + \nu_j^T b \le C^j_{\max} \tag{28}$$

$$\mathrm{CR}^T E_j + \nu_j^T A = 0 \tag{29}$$

$$\nu_j \ge 0 \tag{30}$$

Thus, the RC is as follows:

$$\min_{\mathrm{CR},\mu_j,\nu_j,j\in J} \mathrm{co}^T(\mathrm{ICR} - \mathrm{CR}) \tag{31}$$

$$\text{such that } C^j_{\min} \le a_j^T \mathrm{CR} - \mu_j^T b, \quad \forall\, j \in J \tag{32}$$

$$\mathrm{CR}^T E_j + \mu_j^T A = 0, \quad \forall\, j \in J \tag{33}$$

$$a_j^T \mathrm{CR} + \nu_j^T b \le C^j_{\max}, \quad \forall\, j \in J \tag{34}$$

$$\mathrm{CR}^T E_j + \nu_j^T A = 0, \quad \forall\, j \in J \tag{35}$$

$$\mu_j, \nu_j \ge 0, \quad \forall\, j \in J \tag{36}$$

$$0 \le \mathrm{CR} \le \mathrm{ICR} \tag{37}$$

This problem is convex and can be solved to get an optimal robust solution.

Info-Gap Decision Theory

IGDT entails three components: a system model, uncertainty model, and performance requirement. For illustration, let the system model be $\mathcal{P}$. $\mathcal{P}$ is solved to obtain the optimal decision variable x^*, an optimal value for the parameter ξ^*, and the minimum objective value obj*. The second component is the uncertainty model, which entails the creation of an uncertain set for the uncertain variable, which uses ξ^* as a reference. The performance criterion is dependent on the objective value and a user-specified value for the objective obj. The result of the info-gap decision problem will be an uncertainty set of the specified form (from the uncertainty model) that can be tolerated for the user-specified value objective value obj. As the objective value increases, the uncertain sets will also increase, and the sets will be nested in each other (Ben-Haim 2006).

IGDT Problem Formulation

IGDT does not require an uncertain set, unlike RO; instead, the decision-maker specifies an uncertainty model according to the decision-maker's requirements, whether that may be the reality or not.

The first step in the formulation is the generation of the system model. The system model is the problem $\mathcal{NP}$. $\mathcal{NP}$ is solved assuming complete mixing (i.e., $s = e$) because it is the most popular in the literature. Assuming complete mixing might not give the global least-cost solution for the system model, and the objective value obtained (with $s = e$) is greater than the global least cost. We argue that because this cost is already greater than the global least cost, it can be used in the next stage to develop the uncertainty set for a value of the objective greater than the value obtained because the uncertainty sets are nested. The problem is linear once a value for s is assumed and can be solved to obtain the optimal decision variable CR* and the optimal objective value obj*.

The second component is the uncertainty model. The uncertainty model is assumed to be an envelope-based model, $U(\alpha) = \{s \in \mathbb{R}^I : (1-\alpha\sigma)e \le s \le e\}$, $\alpha \ge 0$ (Ben-Haim 2006), the size of which is controlled by the scalar parameter α. The uncertainty set is a function of α, and as α increases, the set increases in size, but the sets are nested. Because there are bounds on s, that is, $0 \le s \le 1$, correspondingly, the bounds obtained for α are

$$1 - \alpha\sigma \ge 0 \Rightarrow \alpha \le \frac{1}{\sigma} \tag{38}$$

The uncertainty model can be equivalently written as $U(\alpha) = \{s \in \mathbb{R}^I : As \le b\}$, $\alpha \ge 0$ where $A = \begin{bmatrix} I^{I\times I} \\ -I^{I\times I} \end{bmatrix}$ and $b = \begin{pmatrix} e \\ (1-\alpha\sigma)e \end{pmatrix}$, I is the identity matrix, and e is the vector of all 1.

Now, the objective value is allowed to be obj$' >$ obj*, IGDT can be used to develop an uncertainty set and a solution such that the objective value corresponding to the robust solution is at most obj$'$. Thus, the info-gap formulation is as follows:

$$\max \alpha \tag{39}$$

$$\text{such that } \mathrm{co}^T(\mathrm{ICR} - \mathrm{CR}) \le \mathrm{obj}' \tag{40}$$

$$C^j_{\min} \le a_j^T \mathrm{CR} + \mathrm{CR}^T E_j s \le C^j_{\max} \quad \forall\, s \in U(\alpha),\ \forall\, j \in J \tag{41}$$

$$0 \le \mathrm{CR} \le \mathrm{ICR} \tag{42}$$

$$0 \le \alpha \le \frac{1}{\sigma} \tag{43}$$

This formulation [Eqs. (39)–(43)] can be reformulated using the dualization technique, as detailed earlier

$$\max_{\alpha,\mathrm{CR},\mu_j,\nu_j,j\in J} \alpha \tag{44}$$

$$\text{such that } \mathrm{co}^T(\mathrm{ICR} - \mathrm{CR}) \le \mathrm{obj}' \tag{45}$$

$$a_j^T \mathrm{CR} - \mu_j^T b \ge C^j_{\min}, \quad \forall\, j \in J \tag{46}$$

$$\mathrm{CR}^T E_j + \mu_j^T A = 0, \quad \forall\, j \in J \tag{47}$$

$$a_j^T \mathrm{CR} + \nu_j^T b \le C^j_{\max}, \quad \forall\, j \in J \tag{48}$$

$$\mathrm{CR}^T E_j - \nu_j^T A = 0, \quad \forall\, j \in J \tag{49}$$

$$\mu_j, \quad \nu_j \ge 0, \quad \forall\, j \in J \tag{50}$$

$$0 \le \mathrm{CR} \le \mathrm{ICR} \tag{51}$$

$$0 \le \alpha \le \frac{1}{\sigma} \tag{52}$$

where μ_j, $\nu_j J \in J$ is a dual variable that is required for writing the dual problem. The previous formulation gives a larger and larger set as obj$'$ increases because a more robust solution will increase the objective value but will be more immunized against larger realizations of the mixing parameter s. The obtained formulation is nonconvex because of the terms $\mu_j^T b$ and $\nu_j^T b$. The cuckoo search algorithm is used to solve this problem.

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Cuckoo search (CS) is a nature-inspired swarm intelligence algorithm developed by Yang and Deb (2009). This algorithm is based on the breeding behavior of a few species of cuckoo bird. Yang and Deb (2009) stated three basic principles derived from the cuckoo's breeding pattern to convert it to an optimization algorithm. These three rules are: (1) each cuckoo bird lays one egg at a time and dumps it in a randomly chosen nest; (2) the best nests (favorable nests) that resemble the host's eggs are carried to the next time step; and finally (3) the number of available hosts nests are fixed, and any egg laid by a cuckoo bird can be discovered by the host bird with probability $P_a \in [0, 1]$. The working mechanism of the cuckoo search algorithm in (Fig. 2) is extracted from (Pankaj et al. 2020). The first step is to choose the CS algorithm parameters like the number of host nests (n), step size control parameter α_{cs}, and discovery probability (P_a). According to Yang and Deb (2009) $\alpha_{\text{cs}} \cong O\left(\frac{L}{10}\right)$ or $O\left(\frac{L}{100}\right)$ where L is the characteristic scale of the problem of interest, and the discovery probability is assumed to be 0.25. The initial locations of the nests are determined by a set of randomly assigned values within the search space to each decision variable. The initial nests are evaluated, and the best nest is identified based on its fitness value. The local and global searches are performed using Eqs. (53) and (54) in every time step to obtain better solutions.

Local random walk [Eq. (53)] is performed using the P_a parameter, expressed as

$$\text{nest}_i^{\text{new}} = \text{nest}_i^{\text{cur}} + \alpha_{\text{CS}} \otimes H(P_a - \text{rand}) \otimes (\text{nest}_j^{\text{cur}} - \text{nest}_k^{\text{cur}}) \tag{53}$$

where $H(u)$ = Heaviside function; rand = random number generated from a uniform distribution; $\text{nest}_i^{\text{cur}}$, $\text{nest}_j^{\text{cur}}$, and $\text{nest}_k^{\text{cur}}$ = randomly chosen from the current population; and $\otimes$ = entrywise product of two vectors.

A global random walk [Eq. (54)] is performed by using the α_{cs} (step length) and the current best nest using Lévy flight

$$\text{nest}_i^{\text{new}} = \text{nest}_i^{\text{cur}} + \alpha_{\text{cs}} f(\text{best}, \beta_{\text{levy}}) \tag{54}$$

where f = function of the current best nest and levy-flight parameter β_{levy}.

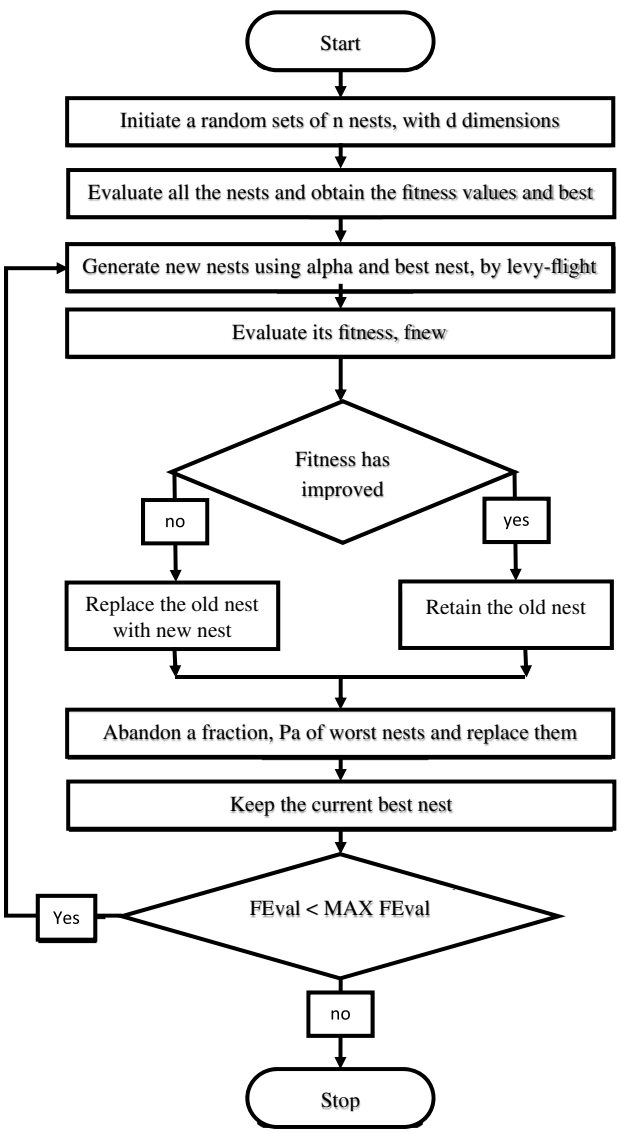

Fig. 2. Flowchart showing the mechanism of the cuckoo search (CS) algorithm.

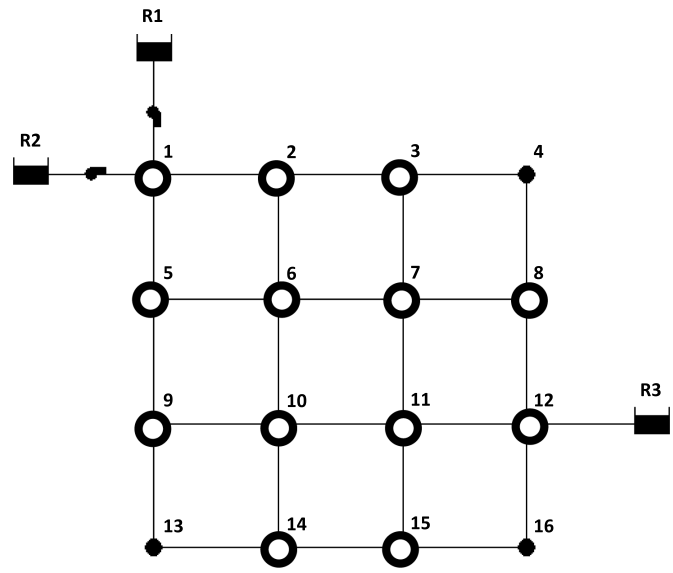

Fig. 3. Study network with all possible junctions where nonuniform mixing can happen (highlighted circles).

Table 1. Illustrative example node data

Node-ID	Base demand (GPM)	Min co-limit	Max co-limit
1	0	0	0
2	10	40	70
3	5	40	70
4	10	40	70
5	50	20	50
6	0	0	0
7	0	0	0
8	10	20	50
9	30	30	60
10	0	0	0
11	0	0	0
12	0	0	0
13	40	30	60
14	80	30	60
15	60	20	50
16	60	40	70

Note: Co-limit = concentration limit; and GPM = gallons per minute.

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Generation and elimination of new nests are performed each time step until the termination criteria are achieved (Yang and Deb 2009). This algorithm and its variations have effectively solved the WDS design problem (Naveen Naidu et al. 2020; Pankaj et al. 2020; Wang et al. 2012). This cuckoo search algorithm is used to obtain the robustness α for the info-gap problem formulation.

Illustrative Example

A 4×4 grid network was considered for this study, as shown in Fig. 3, where all the possible junctions where incomplete mixing is feasible are highlighted. The possibility of incomplete mixing is dependent on flow direction. The objective was to obtain optimal treatment costs. Sources 1 and 2 were at 0 m elevation, and Source 3 was at 70 m. The pump linking Source 1 was of power 2, and the pump linking Source 2 was of power 1.5. The base demand, elevation, and concentration limits of all the nodes are shown in Table 1.

Initial concentrations at the sources before treatment $\mathrm{ICR} = (100, 200, 150)$. Treatment costs for each source $\mathrm{co} = (5{,}000, 200, 10{,}000)$.

Objective function

$$\min_{i=1,2,3} \sum \mathrm{co}^T(\mathrm{ICR} - \mathrm{CR}) \tag{55}$$

where CR = reservoir concentration after treatment entering the network.

Constraints

$$C_j^{\min} \leq C_{\mathrm{out}}^j(\mathrm{CR}, Q, s(\xi))_{\mathrm{BAM}} \leq C_j^{\max} \tag{56}$$

where C_j = concentration of the flow from node j and $C_j^{\min}$ and $C_j^{\max}$ = lower and upper bounds for the concentration at node j.

Surrogate model

As mentioned earlier, to apply RO and IGDT, a surrogate model is needed to replace the BAM model-based equations for outlet concentrations. A unique surrogate equation was developed for each outlet concentration (10 nodes in the illustrative example). After evaluating various forms, the $\mathbf{a_j^T}\mathrm{CR} + \mathrm{CR}^T\mathbf{E_j}s$ form was selected based on the model accuracy. Here, $\mathbf{a_j}$ is a vector, and $\mathbf{E_j}$ is a matrix corresponding to node j. These equations were developed for a particular design of WDN (i.e., constant flow direction and flow values). The procedure to obtain the surrogate model is shown as a flowchart in Fig. 4.

The data generation time for developing the surrogate model for illustrative example was around 14 h and for the modified Fossolo network was around 18 h. The system configuration used for solving the problem was an Intel i5 second-generation processor with 2.1 GHz clock speed and 8 GB RAM

$$C_{\mathrm{out}}^j(\mathrm{CR}, Q, s(\xi))_{\mathrm{BAM}} = a_j^T\mathrm{CR} + \mathrm{CR}^T\mathbf{E}_j\mathbf{s} \tag{57}$$

The surrogate model's accuracy depends on the flow directions and the node location; the maximum RMSE obtained for the cases studied was 2.876, with a minimum $R^2 = 0.989$. This shows that the linear approximation was decently accurate. Monte Carlo

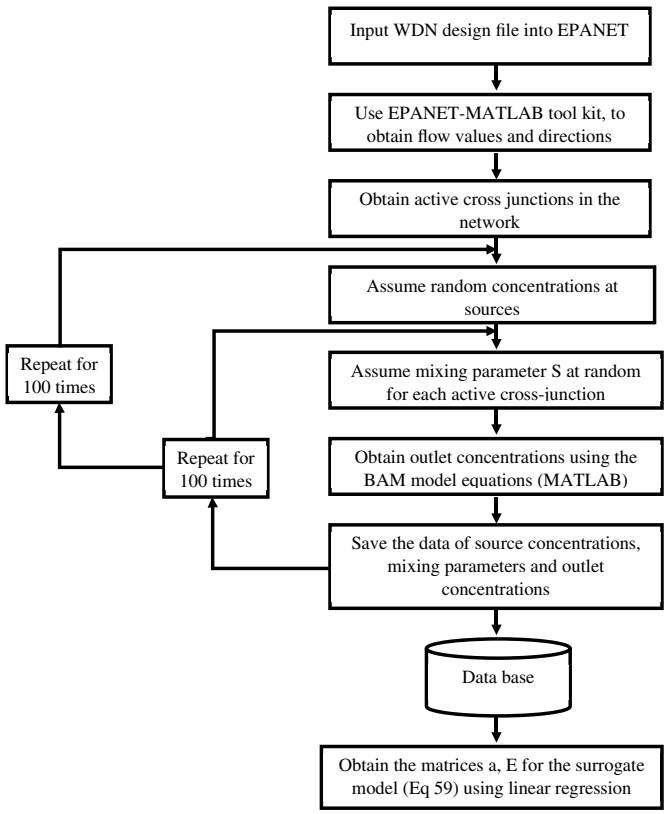

Fig. 4. Surrogate model flow chart.

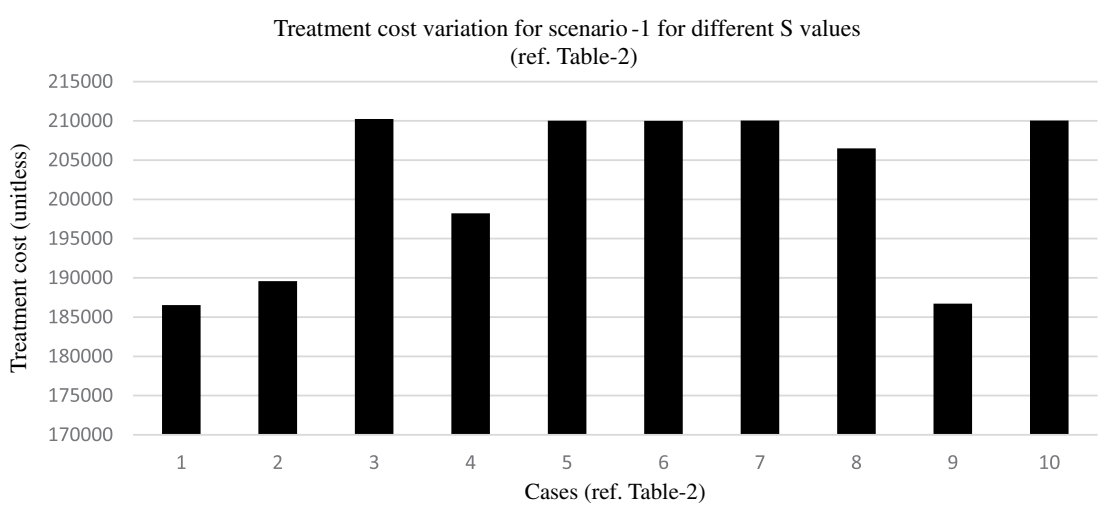

Fig. 5. Variation in the least-cost designs of treatment plants with varying the mixing parameter vector.

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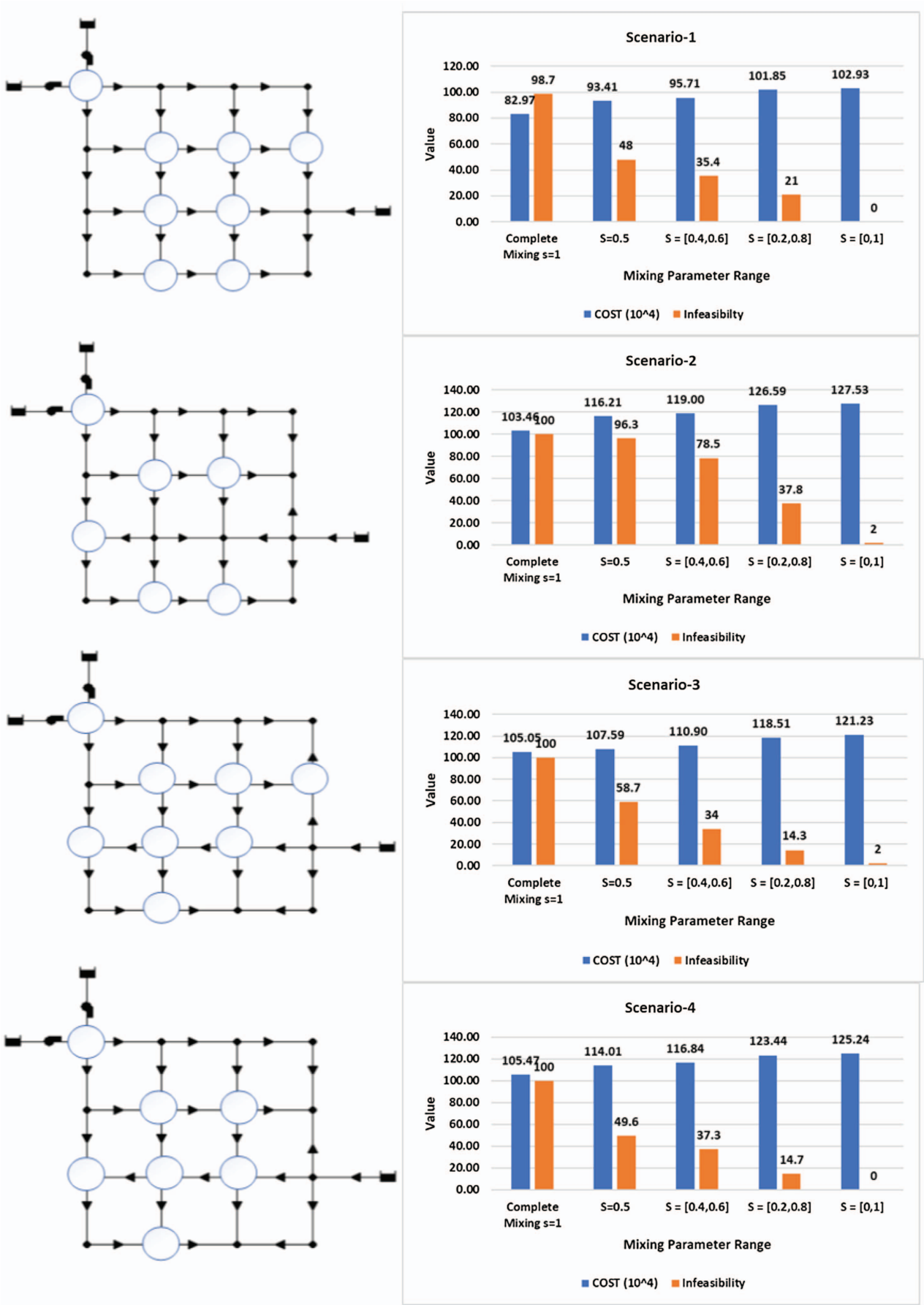

Fig. 6. (Color) Robust optimization results showing the variation in cost for different uncertainty sets.

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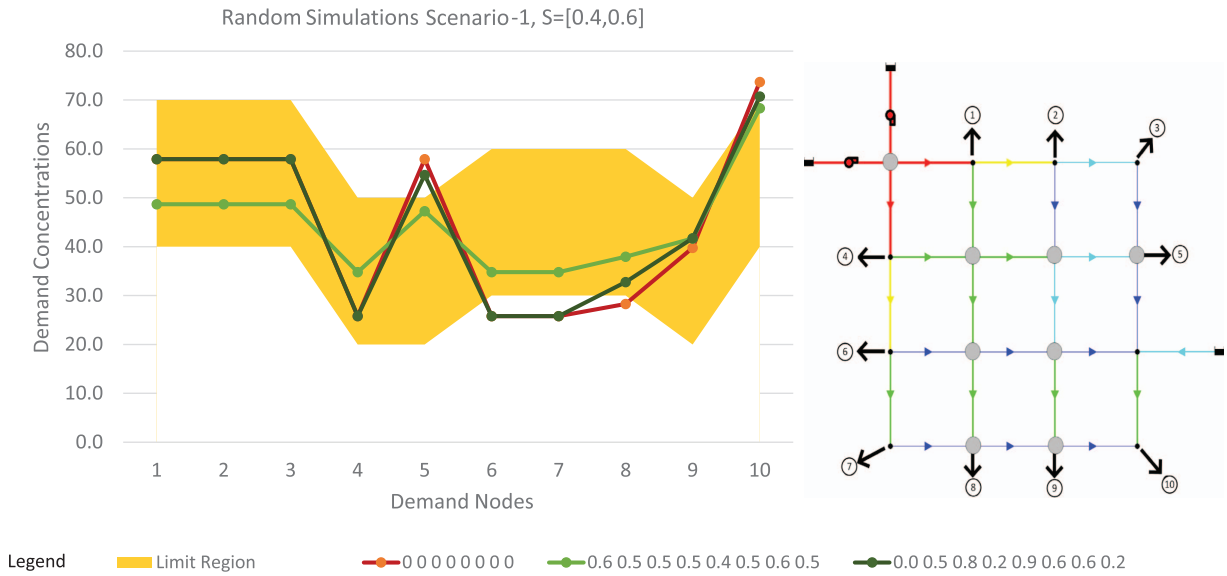

Fig. 7. (Color) Demand concentrations obtained using the optimal concentrations for uncertainty set $s \in [0.4, 0.6]$ under different mixing parameter values.

Table 2. Sensitivity analysis of S vector for Scenario 1

Trial No	Mixing parameter (S) vector	Treatment cost (unitless)
1	**(1, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 1, 0)**	**186,537**
2	(0, 0, 1, 0, 1, 0, 1, 1, 0, 0, 0, 1, 1, 1, 1)	189,587
3	(0, 0, 1, 1, 1, 0, 1, 0, 0, 0, 0, 0, 1, 0, 0)	210,247
4	(0, 1, 0, 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0)	198,217
5	(0.5, 0.2, 0.2, 0.8, 0.2, 0.8, 0.5, 0.2, 0.8, 0.5, 0.5, 0.8, 0.5, 0.5, 0.8)	210,020
6	(0.2, 0.5, 0.8, 0.2, 0.5, 0.2, 0.5, 0.5, 0.8, 0.5, 0.8, 0.5, 0.2, 0.2, 0.8)	210,008
7	**(0.8, 0.8, 0.5, 0.5, 0.2, 0.2, 0.8, 0.5, 0.5, 0.2, 0.5, 0.8, 0.2, 0.2, 0.5)**	**210,040**
8	(0.5, 0.8, 0.8, 0.2, 0.5, 0.8, 0.5, 0.8, 0.2, 0.5, 0.2, 0.2, 0.2, 0.8, 0.5)	206,496
9	**(1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1)**	**186,707**
10	**(0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0)**	**210,051**

Note: Bold values are the lowest treatment cost, highest treatment cost, complete mixing, and incomplete mixing cases.

simulations were used to evaluate the reliability of the obtained design. Sensitivity analysis was done to understand the effect of the mixing parameter on the outlet concentrations (Fig. 5). The same methodology was applied to 10 different flow directions. The obtained results are shown in Figs. 6 and 7. Due to space constraints, results for only four scenarios are reported.

Sensitivity Analysis

A sensitivity analysis was performed to identify how the system model varied with different values of s. The first eight values were chosen randomly, and the last two values represent complete and incomplete mixing, respectively, as stated by Ho et al. (2008). Using these s-values, the optimization problem was solved in a deterministic case, and the corresponding optimal treatment costs are reported in Table 2. Fig. 5 shows how the treatment cost varied with the different values of s.

Fig. 5 and Table 2 show that the variation in optimal treatment costs with the mixing parameters was evident. This variation promoted the need for handling mixing uncertainty in water treatment plant design problems. It can also be inferred that the worst case (i.e., maximum constraint violations) is not easy to estimate. Because the worst case cannot be easily estimated, other nonprobabilistic methods like the min–max method cannot be easily applied to the problem. Thus, efficient nonprobabilistic methods like info-gap decision theory and RO are required to solve the uncertain optimization problem.

Robust Optimization Results

Surrogate models were developed for each scenario (Fig. 6), as explained before. These models were used to convert the uncertain optimization problem to a tractable problem using the RC approach. A box-type uncertainty set was considered. The problem instance contained the flow directions and flow, the treatment cost (co), the surrogate model parameters (a, E), the initial reservoir concentrations (ICR), and the uncertain parameter (s). The decision variable was CR. The optimization problem is shown in Eqs. (10)–(12).

The optimization problem, because s is uncertain, is intractable. The next step is to convert this intractable problem to a tractable form, as detailed in RO methodology. The tractable version of the

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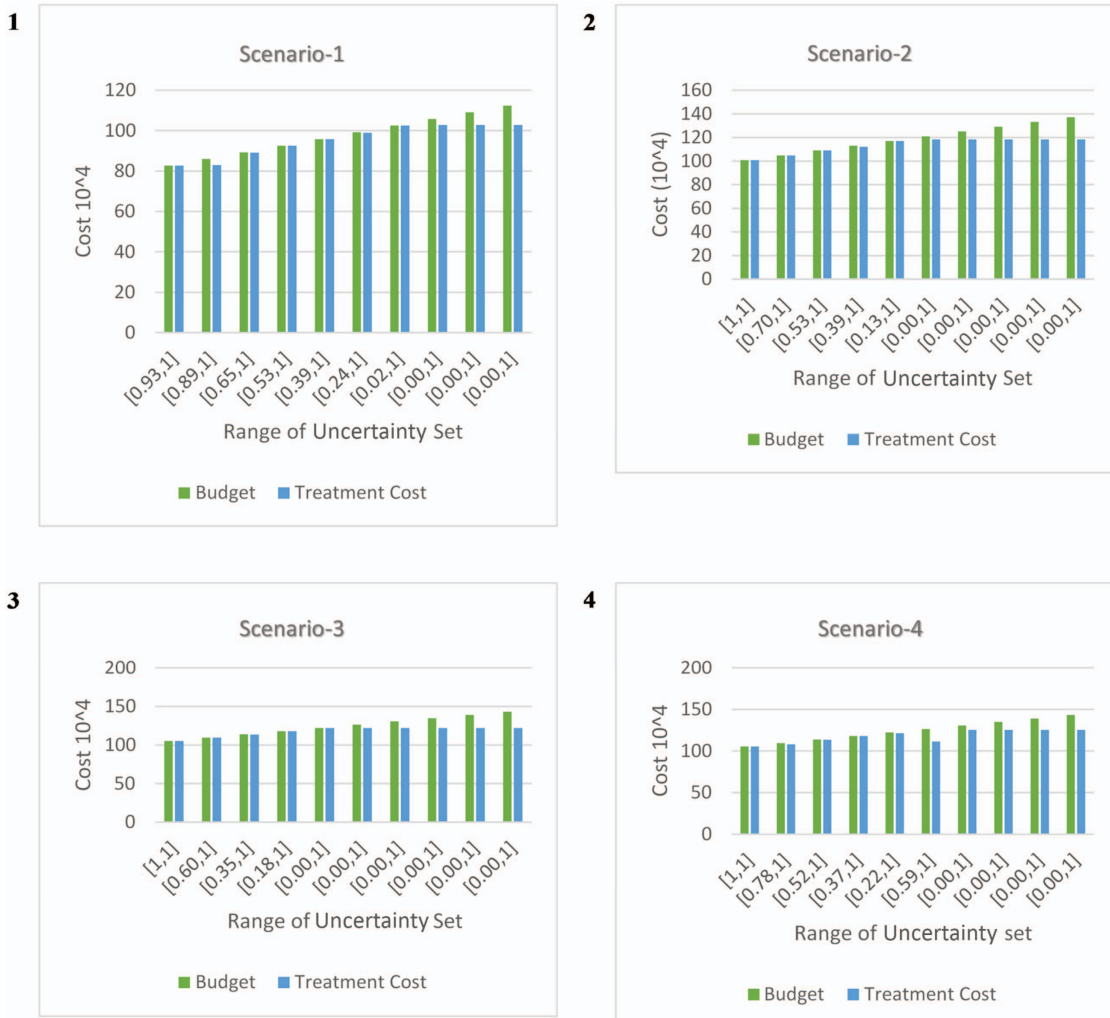

Fig. 8. (Color) Info-gap theory results showing the optimal treatment costs and corresponding robust sets obtained for varying budgets: (1) Scenario 1; (2) Scenario 2; (3) Scenario 3; and (4) Scenario 4.

problem is called the RC, shown in Eqs. (31)–(37), assuming the uncertainty set of s is $s \in [Lb, Ub]$, which can be written as $As \leq b$, $A = \begin{bmatrix} I^{I\times I} \\ -I^{I\times I} \end{bmatrix}$ and $b = \begin{pmatrix} e \times Ub \\ -e \times Lb \end{pmatrix}$. For each scenario, five different uncertainty sets were considered $\{s \in [1, 1], s \in [0.5, 0.5], s \in [0.4, 0.6], s \in [0.2, 0.8], s \in [0, 1]\}$. The optimal treatment costs for each of the uncertainty sets were obtained by solving the RC formulations. The obtained optimal treatment levels were assumed to be attained. The infeasibility percentage was estimated through Monte Carlo simulations by randomly varying the mixing parameter s between $[0, 1]$ and obtaining the outlet concentrations using the original BAM model equations. The percentage of cases where the end constraints are violated was reported as infeasibility percentages. The obtained infeasibility percentages and the related treatment costs are reported in Fig. 6.

The optimal treatment levels obtained for Scenario 1, which were robust to the uncertainty set of $S \in [0.4, 0.6]$, were assumed, and the variation in demand concentrations is shown in Fig. 7. Three cases, (1) incomplete mixing, (2) S within the robust set, and (3) few S values outside the robust set, were considered. The demand concentrations obtained using these mixing parameters are shown in Fig. 7.

Info-Gap Decision Theory Results

After obtaining the surrogate models for each scenario, the nominal or deterministic problem was evaluated assuming complete mixing (i.e., $s = e$) for all cross-junctions. The obtained treatment costs were assumed to be nominal treatment costs and the mixing parameter set ($s = e$) as the nominal mixing parameter value. Using this nominal treatment cost, the maximum budget that could be provided was approximated by linearly increasing the nominal budget (10 times). For each of these approximated budgets, the info-gap problem was solved to obtain the least cost and the corresponding maximum robustness (α). For solving this problem, a cuckoo search optimization algorithm was used. The initial nests (search agents) were taken to be 25, and the Lévy flight parameters $\alpha_{\text{levy}} = 0.01$, $P_a = 0.25$, and $\beta_{\text{levy}} = 2/3$, as suggested in Yang and Deb (2009), are used. The algorithm was run for 200 iterations, or $25 \times 2 \times 200 = 10{,}000$ function evaluations. For all the cases, the problem converged at 100 ± 10 iterations. The uncertainty set obtained from this (α) is reported. For each scenario, 10 different linearly increasing budgets (green) were solved, and their corresponding maximum treatment cost (blue) and robust uncertainty set (x-axis) are reported in Fig. 8.

The optimal treatment levels obtained for Scenario 1, which are robust to the uncertainty set of $S \in [0.65, 1]$, are assumed, and the

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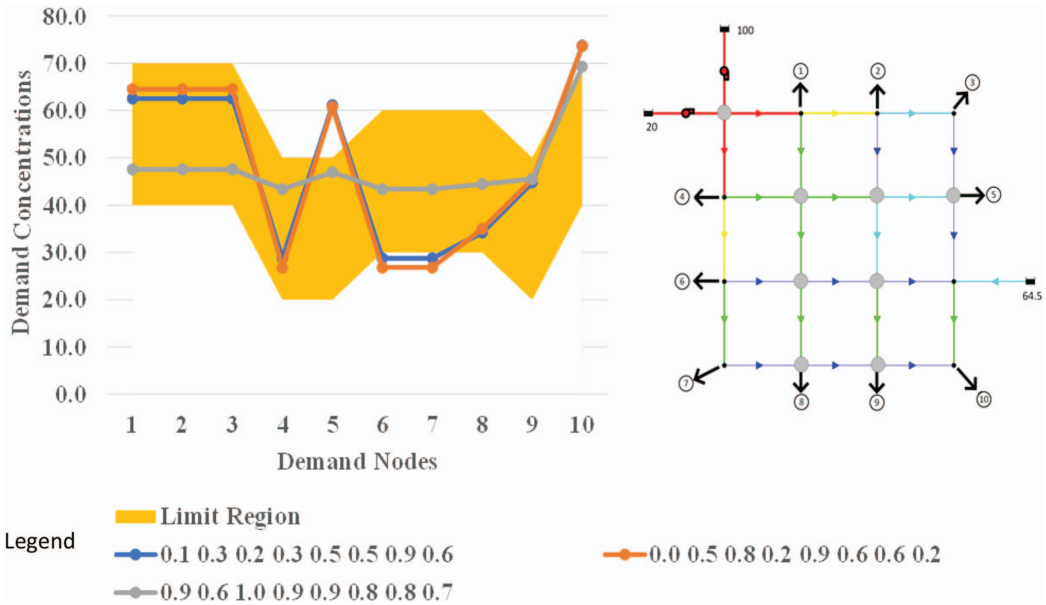

Fig. 9. (Color) Demand concentrations obtained using the optimal concentrations for uncertainty set $s \in [0.65, 1]$ for Scenario 1 under different mixing parameter values.

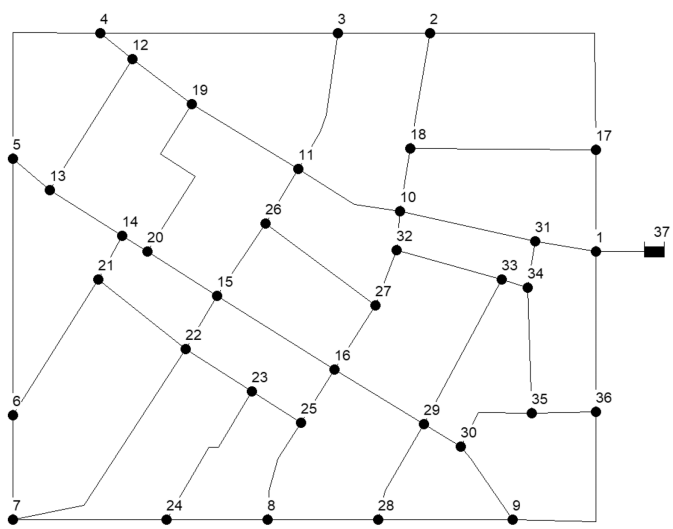

Fig. 10. Fossolo (FOS) network. (Data from Bragalli et al. 2006.)

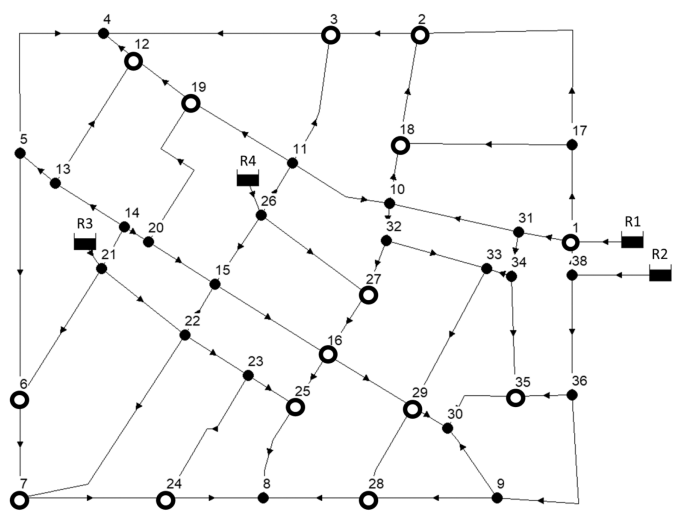

Fig. 11. Modified Fossolo network with four sources. The highlighted nodes have incomplete mixing.

variation in demand concentrations is shown in Fig. 9. Two cases where the mixing parameters are outside the set and one case with mixing parameters were compared. Nodes 5, 6, 7, and 10 were critical for this flow direction. The demand concentrations obtained using these mixing parameters are shown in Fig. 9.

Modified-Fossolo Network

The Fossolo (FOS) network is a real large network in the Fossolo neighborhood of Bologna (Bragalli et al. 2006) on which many authors have applied their design methodologies in the past. The standard FOS has a single source, 58 pipes, and 36 demand nodes. The topology of the network is shown in Fig. 10. This network was chosen because it has many cross-junctions. To increase the complexity, three additional sources were added to the standard FOS network, as shown in Fig. 11. The demands and elevations were modified to create a variation in the flow patterns in the pipes. The modified parameters and the complete details of this modified network are shown in the appendix. This modified FOS network was used for surrogate model generation.

Problem Definition

All four sources were assumed to be contaminated with nonreactive contaminants. The objective of the problem was to find the optimal treatment cost considering nonuniform mixing at cross-junctions. The mathematical formulations are similar to the illustrative example problem.

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Table 3. Concentration limits for nodes of modified FOS

Node ID	C_Max	C_Min	Node ID	C_Max	C_Min	Node ID	C_Max	C_Min
2	50.00	20.00	13	50.00	10.00	27	45.00	10.00
3	50.00	20.00	14	48.00	10.00	28	60.00	10.00
4	50.00	30.00	17	60.00	10.00	30	70.00	20.00
5	70.00	10.00	18	55.00	10.00	31	60.00	30.00
6	70.00	10.00	19	60.00	10.00	32	50.00	15.00
7	70.00	10.00	20	50.00	10.00	33	60.00	34.00
8	60.00	30.00	23	60.00	10.00	34	50.00	25.00
9	70.00	10.00	24	50.00	10.00	35	70.00	30.00
12	55.00	10.00	25	60.00	10.00	36	60.00	10.00

Note: All other nodes have no demand. C_Max = upper limit of outlet concentration; and C_Min = lower outlet concentration limit.

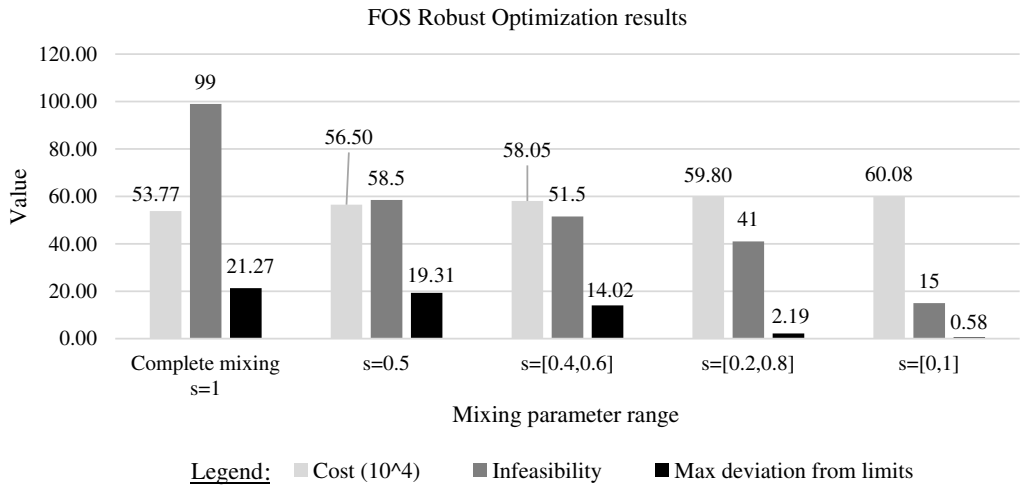

Fig. 12. Robust optimization results for modified FOS network.

Solution Methodology

The problem was solved using both robust optimization and info-gap decision theory methodologies, as explained previously.

The initial concentrations of the sources before the treatment were $\mathrm{ICR} = (100, 100, 100, 100)$, and the corresponding treatment costs for each source were $\mathrm{co} = (500, 2{,}000, 8{,}000, 1{,}500)$. The maximum concentration and minimum limit are shown in Table 3.

Robust Optimization

Similar to the illustrative example, a linear surrogate model [Eq. (57)] was generated and used to formulate the robust counterpart [Eqs. (31)–(37)]. The maximum root mean square error (RMSE) = 0.569 and minimum $R^2 = 0.98$ were the metrics obtained for the surrogate model. Similar to illustrative example, optimal treatment levels were obtained for five different uncertainty sets ($s = 1$, $s = 0.5$, $s = [0.6, 0.4]$, $s = [0.8, 0.2]$, $s = [1, 0]$). The robustness of each design was tested using 1,000 Monte Carlo samples. A graph depicting the cost and their corresponding infeasibility percentages is shown in Fig. 12.

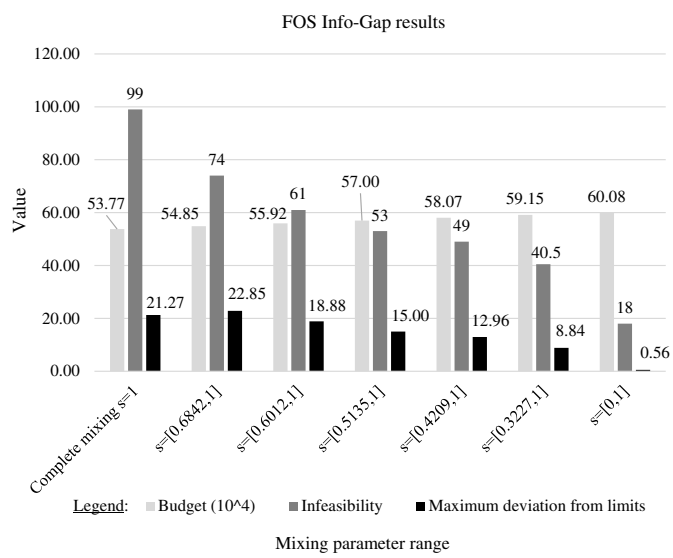

Fig. 13. Info-gap decision theory results for modified FOS network.

Info-Gap Decision Theory

Similar to the illustrative example, first, we found the minimum budget required to satisfy the constraints when there is complete mixing at every junction (i.e., $s = 1$). We then increased the budget and found the maximum uncertainty set of the mixing parameter s was possible within that budget. For our problem, we increased the budget by 2% at each trial and obtained the corresponding uncertainty sets for each budget. Similar to RO, each of the obtained designs was evaluated through 1,000 Monte Carlo simulations. Fig. 13 shows the budget assigned for the designs and their infeasibility.

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Conclusions

The present work was to understand the effect of nonuniform mixing as a water quality uncertain parameter. A simple treatment cost optimization problem was devised to understand the optimal design of nonuniform mixing in a water distribution system. Nonuniform mixing uncertainty cannot assume any probability distribution. Handling this uncertainty requires a nonprobabilistic approach. RO and IGDT nonprobabilistic methods were explored in this study. The complete methodology was explained with a simple grid network with three water sources. The complexity of the nonuniform mixing was clearly explained.

With varying flow patterns, the outlet concentration patterns change. The worst case cannot be assumed because it changes with a change in flow directions and values. Considering incomplete mixing at all junctions did not lead to the worst case (i.e., maximum constraint violations). The change in flow patterns changed the locations of cross-junctions, changing the nonuniform mixing junctions. A new set of surrogate models was required for a change in each flow direction and flow value. A convex surrogate model that includes flow values and direction with reasonable accuracy is difficult to achieve.

The treatment cost increased with an increase in the uncertainty set, as observed from both results. Change in flow direction changed the optimal treatment levels. These similar patterns can also be inferred from the info-gap results. The Monte Carlo simulation results showed a small percentage of infeasibility was due to the surrogate model's approximation. Both approaches can be easily applied to any network and solve this optimal treatment problem. Further study would be to combine the water distribution network design problem with this treatment cost design problem.

The proposed robust optimization methodology was applied to a real network, the Fossolo network. Even for the FOS network, a similar surrogate model provided a decent approximation. The obtained results were tested to check the probability of infeasibility through Monte Carlo simulations. This showed that the proposed methodology can be applied to any network. The only drawback was that, to obtain the linear surrogate model, the computational time to generate the data was directly proportional to the number of nodes.

It should be emphasized that the particular challenges in this study were in developing the proposed RO and info-gap methods and demonstrating them through base runs and sensitivity analyses on two example applications from the research literature. As such, the emphasis was more on the theory and less on using real data. This is why unitless cost data and conservative water quality constituents were used. Applications to nonconservative water quality parameters, real cost data, and more complex systems are suggested as extensions to this work.

Data Availability Statement

All data, models, or codes that support this study's findings are available from the corresponding author upon reasonable request.

Acknowledgments

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